

July 11, 2025

Lapidot: Chapter 2 Signals, integrals and sets
of measure zero.

2.8 Exercises

2.1 (Scaling the Integral) Starting from the definition of the integral of a complex signal (2.8) prove that if $u: \mathbb{R} \rightarrow \mathbb{C}$ is integrable and a is a complex number, then

$$\int_{-b}^{+b} a u(t) dt = a \int_{-b}^{+b} u(t) dt$$

Solution:

$$\begin{aligned} \int_{-b}^{+b} a u(t) dt &= \int_{-b}^{+b} \operatorname{Re}\{a u(t)\} dt \\ &\quad + i \int_{-b}^{+b} \operatorname{Im}\{a u(t)\} dt. \end{aligned}$$

$$\begin{aligned} &\operatorname{Re}\{(a+ib)(c+id)\} \\ &= \operatorname{Re}\{ac + iad + ibc - bd\} \\ &= \operatorname{Re}\{ac - bd + i(ad + bc)\} \\ &= ac - bd = \operatorname{Re}\{x\} \operatorname{Re}\{y\} - \operatorname{Im}\{x\} \operatorname{Im}\{y\}. \end{aligned}$$

(2)

$$\operatorname{Im} \{(a+ib)(c+id)\}.$$

$$= \operatorname{Im} \{ac - bd + i(ad+bc)\}$$

$$= ad+bc = \operatorname{Re}\{x\}\operatorname{Im}\{y\} + \operatorname{Im}\{x\}\operatorname{Re}\{y\}.$$

$$\begin{aligned} \text{Thus, } \int_{-t_0}^{t_0} a u(t) dt &= \int_{-t_0}^{t_0} \operatorname{Re}\{a\} \operatorname{Re}\{u(t)\} - \operatorname{Im}\{a\} \operatorname{Im}\{u(t)\} dt \\ &\quad + i \int_{-t_0}^{t_0} \operatorname{Re}\{a\} \operatorname{Im}\{u(t)\} + \operatorname{Im}\{a\} \operatorname{Re}\{u(t)\} dt \\ &= \frac{a'}{c'} \int_{-t_0}^{t_0} \operatorname{Re}\{u(t)\} dt - \frac{b'}{d'} \int_{-t_0}^{t_0} \operatorname{Im}\{u(t)\} dt \\ &\quad + i \left[\frac{\operatorname{Re}\{a\}}{a'} \int_{-t_0}^{t_0} \operatorname{Im}\{u(t)\} dt + \frac{\operatorname{Im}\{a\}}{b'} \int_{-t_0}^{t_0} \operatorname{Re}\{u(t)\} dt \right] \\ &= (a'c' - b'd') + i(a'd' + b'c') \\ &= (a' + ib')(c' + id') \\ &= a \int_{-t_0}^{t_0} u(t) dt \quad \square. \end{aligned}$$

Exercise 2.2. (A useful change-of-variable) Let

x be an integrable function, and let $\bar{x}$ be its mirror image. Prove that:

$$\int_{-b}^{+b} \overleftarrow{x}(t) dt = \int_{-b}^{+b} x(t) dt$$

③

$$\overleftarrow{x}(t) = x(-t)$$

$$\begin{aligned} \int_{-b}^{+b} \overleftarrow{x}(t) dt &= \int_{-b}^{+b} x(-t) dt \\ &= \int_{+b}^{-b} x(u) - du \\ &= - \int_{+b}^{-b} x(u) du \\ &= \int_{-b}^{+b} x(u) du = \int_{-b}^{+b} x(t) dt. \end{aligned}$$

EXERCISE 2.3 (Integrating a complex integrand).

$$\underline{\text{P.T.}} \quad \int_0^{\infty} e^{-zt} dt = \frac{1}{z}, \quad \operatorname{Re}(z) > 0.$$

This is a complex integrand. $\underline{u}: \mathbb{R} \rightarrow \mathbb{C}$

We use the following definition of a complex integral given

in the book:

$$\int_{-b}^{+b} u(t) dt = \int_{-b}^{+b} \operatorname{Re}\{u(t)\} dt + i \int_{-b}^{+b} \operatorname{Im}\{u(t)\} dt$$

For complex z , Euler's formula comes into play:

(4)

$$e^{-zt} = e^{-(u+iv)t} = e^{-ut} \cdot e^{-ivt} \\ = e^{-ut} \cos t - i e^{-ut} \sin t$$

$$\operatorname{Re}\{z\} > 0 \Rightarrow u > 0.$$

$$\text{Thus } \int_0^\infty e^{-zt} dt = \int_0^\infty e^{-ut} \cos t dt - i \int_0^\infty e^{-ut} \sin t dt$$

$$\begin{aligned} & \left(\begin{array}{l} \text{using} \\ \text{Mathematica/} \\ \text{Integration} \\ \text{tables.} \end{array} \right) = \frac{u}{u^2+v^2} - \frac{iv}{u^2+v^2} \\ & = \frac{z^*}{|z|^2} = \frac{1}{2} \cdot \frac{z^*}{z^*} = \frac{1}{z} \quad \square. \end{aligned}$$

EXERCISE 2.4 (The Sine is not Integrable) Show that the

mapping $t \mapsto \frac{\sin(\pi t)}{\pi t}$ is not integrable, irrespective of how we define it at zero.

Hint: lower bound the integral of the absolute value of this function from n to $n+1$ by the integral from $n+\frac{1}{4}$ to $n+\frac{3}{4}$. Lower bound the integrand

(5)

in this region, and use the fact the harmonic series

$\sum_{k=0}^{\infty} \frac{1}{k}$ diverges.

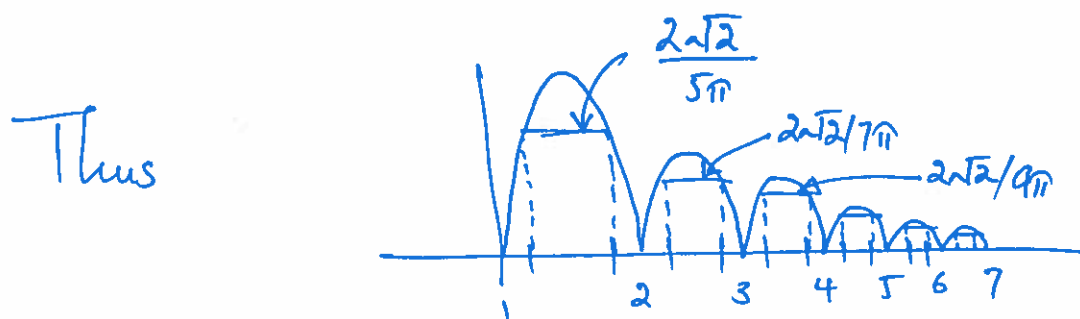
Solution: This is a simple exercise in lower bounding

the area given by:
$$A = \int_1^{\infty} \left| \frac{\sin(\pi t)}{\pi t} \right| dt$$

$$= \sum_{n=1}^{\infty} \int_n^{n+1} \left| \frac{\sin(\pi t)}{\pi t} \right| dt$$

Plot it.
$$\textcircled{1} \longrightarrow \geq \sum_{n=1}^{\infty} \int_{n+\frac{1}{4}}^{n+\frac{3}{4}} \left| \frac{\sin(\pi t)}{\pi t} \right| dt$$

$$\left\{ \left[n + \frac{1}{4}, n + \frac{3}{4} \right] : [1.25, 1.75], [2.25, 2.75], \dots \right\}$$



Because
$$\frac{\sin\left[\pi\left(n+\frac{1}{4}\right)\right]}{\pi\left(n+\frac{1}{4}\right)}, \quad n=1, 2, 3, 4, \dots$$

$$= \frac{2\sqrt{2}}{\pi(2n+1)} \quad (\text{Check using Mathematica})$$

Thus
$$\textcircled{1} \geq \sum_{n=1}^{\infty} \frac{1}{2} \cdot \frac{2\sqrt{2}}{\pi(2n+1)}$$

$$= \frac{\sqrt{2}}{\pi} \sum_{n=1}^{\infty} \frac{1}{(2n+1)}$$

$$\geq \frac{\sqrt{2}}{\pi} \sum_{n=1}^{\infty} \frac{1}{(2n+2)}$$

$$= \frac{\sqrt{2}}{2\pi} \sum_{n=1}^{\infty} \frac{1}{n+1}$$

$$= \frac{\sqrt{2}}{2\pi} \left[\underbrace{\sum_{n=2}^{\infty} \frac{1}{n}}_{=\infty} + \frac{1}{2} \right].$$

Thus $\int_0^{\infty} \frac{\sin(\pi t)}{\pi t} dt$ is not integrable and

since $\int_{-\infty}^{+\infty} \frac{\sin(\pi t)}{\pi t} dt = 2 \int_0^{\infty} \frac{\sin(\pi t)}{\pi t} dt$

↓

~~∞~~ as is not integrable as well $\square$.

EXERCISE 2.5: Prove the triangular inequality for complex numbers. Under what condition does it hold with equality.

Solution: There is an easy way to do this and a hard way to do this. We will follow the easy way.

Identify $\mathbb{C} \rightarrow \mathbb{R}^2$

$$a+ib \longleftrightarrow (a,b)$$

7

$$|\underline{u} + \underline{v}| \leq |\underline{u}| + |\underline{v}|$$

$$\sqrt{(a+c)^2 + (b+d)^2} \leq \sqrt{a^2+b^2} + \sqrt{c^2+d^2}$$

$$(a+c)^2 + (b+d)^2 \leq a^2+b^2 + c^2+d^2 + 2\sqrt{(a^2+b^2)(c^2+d^2)}$$

$$a^2+c^2+b^2+d^2 + 2ac + 2bd \leq 2\sqrt{(a^2+b^2)(c^2+d^2)} + a^2+b^2+c^2+d^2$$

$$2ac + 2bd \leq 2\sqrt{(a^2+b^2)(c^2+d^2)}$$

$$\underline{\underline{ac + bd \leq \sqrt{(a^2+b^2)(c^2+d^2)}}}$$

Cauchy-Schwarz.

$$\left| \sum u_i v_i \right| \leq \sqrt{\left(\sum u_i^2 \right) \left(\sum v_i^2 \right)}$$

or

$$|\underline{u} \cdot \underline{v}| \leq |\underline{u}| |\underline{v}|.$$

Equality is achieved if $\underline{u} = \alpha \underline{v}$ $\alpha \in \mathbb{C}$.

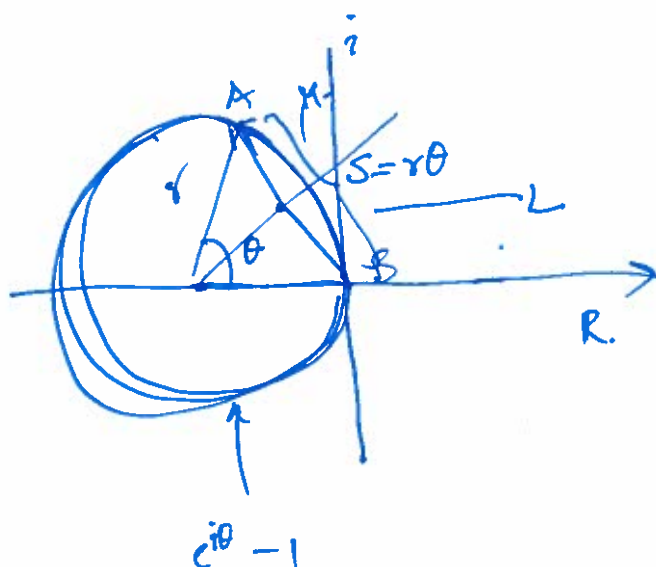
EXERCISE 2.7 (Bounding Complex Exponentials)

(8)

Show that:

$$|e^{i\theta} - 1| \leq |\theta|, \quad \theta \in \mathbb{R}.$$

Solution: The proof is geometric rather than analytic (not calculus based)



$e^{i\theta} - 1$ is the unit circle displaced left by 1 unit.

The point $e^{i\theta} - 1$ is the locus of points of the circle and $|e^{i\theta} - 1| = L$ is the length of the chord having one point at $(1, 0)$ and the other at $e^{i\theta} - 1$. From the geometry above we immediately see that

$$\frac{1}{r} \frac{L}{2} = \left| \sin\left(\frac{\theta}{2}\right) \right|$$

$$L = \left| 2r \sin\left(\frac{\theta}{2}\right) \right| = 2 \left| \sin\left(\frac{\theta}{2}\right) \right| \quad \text{since } r=1$$

$$\text{Thus, } |e^{j\theta} - 1| = L = \left| 2 \sin\left(\frac{\theta}{2}\right) \right| \leq 2 \left| \frac{\theta}{2} \right|$$

$$= |\theta| \quad \square.$$

EXERCISE 2.8. : (An Integral Inequality)

Show that if u , v and w are integrable signals then

$$\int_{-\infty}^{+\infty} |u(t) - w(t)| dt \leq \int_{-\infty}^{+\infty} |u(t) - v(t)| dt + \int_{-\infty}^{+\infty} |v(t) - w(t)| dt$$

Proof: We use the fact if $x(t) \geq 0 \Rightarrow \int_{-\infty}^{+\infty} x(t) dt \geq 0$.

$$\text{So if } y(t) \geq x(t) \Rightarrow \int x(t) dt \leq \int y(t) dt.$$

$$\text{Thus, } |u(t) - w(t)| = |u(t) - v(t) + v(t) - w(t)|$$

$$\leq |u(t) - v(t)| + |v(t) - w(t)|$$

$$\text{And so, } \int_{-\infty}^{+\infty} |u(t) - w(t)| dt \leq \int_{-\infty}^{+\infty} |u(t) - v(t)| dt$$

$$+ \int_{-\infty}^{+\infty} |v(t) - w(t)| dt \quad \square.$$

EXERCISE 2.9: (An Integral to Note): $f \in \mathbb{R}$

compute the integral

$$\int_{-\infty}^{+\infty} I\{t=17\} e^{-j2\pi f t} dt.$$

Proof: Some confusion: is $I\{t=17\} \equiv \delta(t-17)$

$$\begin{aligned} \text{then } \int_{-\infty}^{+\infty} \delta(t-17) e^{-j2\pi f t} dt &= e^{-j2\pi f 17} \\ &= e^{-j14\pi f} \end{aligned}$$

$$\text{Otherwise: } \int I(t=17) e^{-j2\pi f t} dt = 0$$

$$\text{because } \mu(\{t: I(t=17) \neq 0\}) = 0.$$

EXERCISE 2.10 Show that (subsets of sets of measure

zero) of a set of measure zero $\nRightarrow$ has measure zero.

(Measure $\equiv$ Lebesgue measure).

Let $A \subseteq B$. $\mu(B) = 0$.

⑨

$$\Rightarrow \exists \{[a_i, b_i] : B \subseteq \bigcup_i [a_i, b_i], \sum_i (b_i - a_i) < \epsilon, \forall \epsilon > 0\}.$$

Since, $A \subseteq B \Rightarrow A \subseteq \bigcup_i [a_i, b_i]$

with $\sum_i (b_i - a_i) < \epsilon, \forall \epsilon > 0$

$$\Rightarrow \mu(A) = 0.$$

EXERCISE: 2.11 (The union of Sets of Lebesgue measure zero)

Show that the union of two sets that are each of Lebesgue measure zero is also of Lebesgue measure zero.

Given.

Solution:

$$\exists [a_i, b_i] : A \subseteq \bigcup_i [a_i, b_i] \mid \forall \epsilon > 0 \sum_i (b_i - a_i) < \epsilon$$

$$\exists [c_i, d_i] : B \subseteq \bigcup_i [c_i, d_i] \mid \forall \epsilon > 0 \sum_i (d_i - c_i) < \epsilon$$

$$A \cup B \subseteq \bigcup_i [a_i, b_i] \cup \bigcup_i [c_i, d_i]$$

$$\text{and } \sum_i (b_i - a_i) < \epsilon/2 \wedge \sum_i (d_i - c_i) < \epsilon/2$$

$$\Rightarrow \sum_i (b'_i - a'_i) = \sum_i (b_i - a_i) + \sum_i (d_i - c_i) < \epsilon/2 + \epsilon/2 = \epsilon.$$

EXERCISE 2.12 (Non-uniqueness of pdf) We say that (12)

the random variable $\underline{X}$ is of density $f_{\underline{X}}(\cdot)$ if $f_{\underline{X}}(\cdot)$ is a (Lebesgue measurable) non-negative function such

that:

$$P(\underline{X} \leq x) = \int_{-\infty}^x f_{\underline{X}}(\xi) d\xi$$

Show that if $\underline{X}$ has a density $f_{\underline{X}}(\cdot)$ and if $g(\cdot)$ is a non-negative function that is indistinguishable from $f_{\underline{X}}(\cdot)$, the $\underline{X}$ also has density $g(\cdot)$.

Solution: This is a standard result in Lebesgue theory.

We have $\mu(\{t: f_{\underline{X}}(t) \neq g(t), t \in \mathbb{R}\}) = 0$

then $\int_{-\infty}^{+\infty} |f_{\underline{X}}(t) - g(t)| dt = 0$ (Proposition 2.5.3 (page 9))

$$\geq \left| \int_{-\infty}^{+\infty} f_{\underline{X}}(t) - g(t) dt \right| \geq 0$$

$$\Rightarrow \left| \int_{-\infty}^{+\infty} (f_{\underline{X}}(t) - g(t)) dt \right| = 0$$

$$\Rightarrow \int (f_{\underline{X}} - g) = 0 \Rightarrow \int f_{\underline{X}} = \int g$$

Thus,

$$P^f[\underline{x} \leq x] = P^g[\underline{x} \leq x].$$

(13)

□.

EXERCISE 2.13 (Indistinguishability). Let $\psi: \mathbb{R}^2 \rightarrow \mathbb{R}$

satisfy $\psi(\alpha, \beta) \geq 0$ for $\forall \alpha, \beta \in \mathbb{R}$ with equality if and

only if $\alpha = \beta$. Let $\underline{u}, \underline{v}$ be Lebesgue measurable signals.

Show that: $\int_{-\infty}^{\infty} \psi(u(t), v(t)) dt = 0 \Rightarrow \underline{u} \equiv \underline{v}$

Consider the set

$$E = \{t: \psi(u(t), v(t)) > 0\}.$$

$$= \{t: u(t) \neq v(t)\}$$

Assume ψ is Lebesgue integrable.

$$\int_{-\infty}^{+\infty} \psi(u(t), v(t)) dt = 0$$

$\Rightarrow \psi(u(t), v(t))$ is indistinguishable from zero.

$$\Rightarrow \mu(\{t: \psi(u(t), v(t)) \neq 0\}) = 0$$

$$\Rightarrow \mu(\{t: u(t) \neq v(t)\}) = 0 \Rightarrow \underline{u} \equiv \underline{v} \quad \square.$$

EXERCISE 2.14 : (Indistinguishable signals). Show that

(14)

if the Lebesgue measurable signal $\underline{g}$ and $\underline{h}$ are indistinguishable then the set of epochs $t \in \mathbb{R}$ where the sums $\sum_{j=-\infty}^{+\infty} g(t+j)$ and $\sum_{j=-\infty}^{\infty} h(t+j)$ are different (in the sense that they converge but to different limits or that one converges but the other does not) is of Lebesgue measure zero.

Solution: We will prove the special case of both sums converging. We will use Fubini + Markov Inequality. Define,

$$F = \left\{ t : \left| \sum_j (h(t+j) - g(t+j)) \right| > 0 \right\} \quad \Bigg| \rightarrow t \in \mathbb{R}.$$
$$E = \left\{ t : \sum_j |h(t+j) - g(t+j)| > 0 \right\}.$$

$$\text{Since } \left| \sum_j h(t+j) - \sum_j g(t+j) \right| \leq \left| \sum_j (h(t+j) - g(t+j)) \right|$$
$$\leq \sum_j |h(t+j) - g(t+j)|$$

We have $F \subseteq E$. So we only need to show

(15)

that $\mu(E) = 0$ (Since μ is a complete measure).

Define, $E_n = \left\{ t: \sum_j |h(t+j) - g(t+j)| > \frac{1}{n} \right\}$

Then, $\mu(E_n) \leq n \cdot \int_{-n}^{+n} \sum_j |h(t+j) - g(t+j)| d\mu$.
Markov

(Fubini) $= n \cdot \sum_j \int_{-n}^{+n} |h(t+j) - g(t+j)| dt$

$\underline{h} \equiv \underline{g}$ $= n \cdot \sum_j 0 = 0$

Thus $\mu(E) = \mu\left(\bigcup_n E_n\right) \leq \sum \mu(E_n) = 0$.

Thus $\mu\left(\left\{ t: \sum h(t+j) \neq \sum g(t+j) \right\}\right) = 0$.
 $\square$

2.15 EXERCISE: (Continuous Non-negative Functions).

A subset $\mathbb{R}$ containing a non-empty open interval

cannot be a Lebesgue measure zero. Use this

fact to show that if a continuous function $\underline{g}: \mathbb{R} \rightarrow \mathbb{R}$

is non-negative except perhaps on a set of Lebesgue (16)
measure zero, then the exception set is empty and the
function is non-negative.

Solution: This is an easy exercise in basic analysis.

if $f \in C^0$ and $f \geq 0$ with $\mu(\{f < 0\}) = 0$.

Now, let $E = \{t: f(t) < 0\}$. ^{Assume $E \neq \emptyset$} $\Rightarrow \exists t_0 \in E$ such that

$f(t_0) < 0 \Rightarrow \exists \epsilon > 0$ $(t_0 - \epsilon, t_0 + \epsilon) \subseteq E$ because

f is continuous. Thus, $\mu(E) \geq 2\epsilon > 0$. But this is

is a contradiction. $\Rightarrow E = \emptyset$ and the function is

purely non-negative

□.

EXERCISE 2.16 (Order of summation sometimes matters).

$\nu, \eta \in \mathbb{N}$ define,

$$a_{v,\eta} = \begin{cases} 2 - 2^{-v} & \text{if } v = \eta \\ -2 + 2^{-v} & \text{if } v = \eta + 1 \\ 0 & \text{else.} \end{cases}$$

Show that, (2.21) is not satisfied. (from Royden)

$$(2.21) \rightarrow \sum_{v=1}^{\infty} \left(\sum_{\eta=1}^{\infty} a_{v,\eta} \right) = \sum_{\eta=1}^{\infty} \left(\sum_{v=1}^{\eta} a_{\eta,v} \right).$$

Solution:

$$\textcircled{1} \quad \sum_{v=1}^{\infty} \left(\sum_{\eta=1}^{\infty} a_{v,\eta} \right) = \sum_{v=1}^{\infty} \left(\cancel{2} - 2^{-v} + \cancel{(-2 + 2^{-(v-1)})} \right)$$

$$= \sum_{v=1}^{\infty} (-2^{-v}) + \sum_{v=2}^{\infty} 2^{-(v-1)}$$

$$= - \sum_{v=1}^{\infty} 2^{-v} + \sum_{v=2}^{\infty} 2^{-v+1}$$

$$= -1 + \sum_{v=2}^{\infty} 2^{-v} \cdot 2$$

$$= -1 + \frac{1}{2} \cdot 2 = -1 + 1 = 0.$$

$$\textcircled{2} \quad \sum_{\eta=1}^{\infty} \left(\sum_{v=1}^{\infty} a_{v,\eta} \right) = \sum_{\eta=1}^{\infty} \left(\cancel{2} - \bar{2}^{-\eta} + \cancel{(-2 + 2^{-(\eta+1)})} \right)$$

$$= - \sum_{\eta=1}^{\infty} \bar{2}^{-\eta} + \sum_{\eta=1}^{\infty} \bar{2}^{\eta+1}$$

$$= -1 + \frac{1}{2} \sum_{\eta=1} \bar{2}^{\eta}$$

$$= -1 + \frac{1}{2} \cdot 1 = -\frac{1}{2}$$

Thus ① $\neq$ ② (2.21) does not hold.

□

EXERCISE 2.17 (Using Fubini's Theorem) Using the relation

$$\frac{1}{x} = \int_0^{\infty} e^{-xt} dt, \quad x > 0$$

and Fubini's Theorem, show that

$$\lim_{\alpha \rightarrow \infty} \int_0^{\alpha} \frac{\sin x}{x} dx = \frac{\pi}{2}.$$

(Rudin 1987 Chapter 8, Exercise 12).

Solution: So let's try a brute force, naive approach.

$$\lim_{\alpha \rightarrow \infty} \int_0^{\alpha} \frac{\sin x}{x} dx = \lim_{\alpha \rightarrow \infty} \int_0^{\alpha} \left(\int_0^{\infty} e^{-xt} dt \right) \sin x dx$$

$$= \lim_{\alpha \rightarrow \infty} \int_0^{\infty} \int_0^{\alpha} e^{-xt} \sin x \, dx \, dt$$

$$= \lim_{\alpha \rightarrow \infty} \int_0^{\infty} \frac{1 - e^{\alpha t} \cos(\alpha) - t e^{\alpha t} \sin(\alpha)}{1+t^2} \, dt$$

$$= \int_0^{\infty} \lim_{\alpha \rightarrow \infty} \frac{1 - e^{\alpha t} \cos \alpha - t e^{\alpha t} \sin \alpha}{1+t^2} \, dt$$

$$= \int_0^{\infty} \frac{1}{1+t^2} \, dt$$

$$= \text{Arctan}(t) \Big|_0^{\infty}$$

$$= \text{Arctan}(\infty) - \text{Arctan}(0)$$

$$= \frac{\pi}{2}.$$

□.

Chapter 3: Inner Product: (page 14)

July 12, 2025

$$\langle \underline{u}, \underline{v} \rangle = \int_{-\infty}^{+\infty} u(t) v^*(t) dt.$$

$$\|\underline{u}\|_2 = \sqrt{\int_{-\infty}^{+\infty} u^2(t) dt}.$$

$$\|\underline{u}\|_2^2 = \int_{-\infty}^{+\infty} u^2(t) dt = \langle \underline{u}, \underline{u} \rangle$$

$$\underline{u}, \underline{v} : \mathbb{R} \rightarrow \mathbb{C}$$

$$\langle \underline{u}, \underline{v} \rangle = \int_{-\infty}^{+\infty} \operatorname{Re}\{u(t) v^*(t)\} dt + i \int_{-\infty}^{+\infty} \operatorname{Im}\{u(t) v^*(t)\} dt$$

$$E_u = \int_{\mathbb{R}} |u(t)|^2 dt$$

DEFINITION 3.1.1. (Energy Limited Signals)

$$E_u = \int |u(t)|^2 dt < \infty.$$

SECTION 3.2: When is an inner product defined?

$\langle \underline{u}, \underline{v} \rangle$ makes sense when?

$\underline{u} : \mathbb{R} \rightarrow \mathbb{C}$ is integrable iff $\rightarrow$

$t \mapsto \operatorname{Re}\{u(t)\}$ and $t \mapsto \operatorname{Im}\{u(t)\}$ are Lebesgue measurable real valued signals and $\int_{-\infty}^{+\infty} |u(t)| dt < \infty$.

Thus $\langle u, v \rangle$ exists $\iff u: \mathbb{R} \rightarrow \mathbb{C} \wedge v: \mathbb{R} \rightarrow \mathbb{C}$ is both Lebesgue measurable. ($\Rightarrow t \mapsto u(t)v^*(t)$ is Lebesgue measurable)

and when:

$$\int_{-\infty}^{+\infty} |u(t)v^*(t)| dt < \infty. \quad \text{--- (3.13)}$$

QUESTION: When do these conditions hold. (3.13) holds.

— Simplest case: u is bounded $\wedge v$ is integrable.

$$|u(t)v(t)| \leq \sigma_\infty |v(t)|$$

$$\begin{aligned} \int_{\mathbb{R}} |u(t)v(t)| dt &\leq \sigma_\infty \int_{-\infty}^{+\infty} |v(t)| dt \\ &= \sigma_\infty \|v\|_1. \end{aligned}$$

— Another case is when both u and v are of finite energy ($E_u, E_v < \infty$)

$$\text{we have} \quad \alpha\beta \leq \frac{1}{2}(\alpha^2 + \beta^2)$$

We thus have:

$$\int_{-\infty}^{+\infty} |u(t)v(t)| dt \leq \frac{1}{2} \left(\int_{-\infty}^{+\infty} |u(t)|^2 dt + \int_{-\infty}^{+\infty} |v(t)|^2 dt \right) < \infty.$$

— As a by product of $\alpha\beta \leq \frac{1}{2}(\alpha^2 + \beta^2)$

We can upper bound the inner product by the energies of u and v . Note:

$$\left| \int_{-\infty}^{+\infty} f(\xi) d\xi \right| \leq \int_{-\infty}^{+\infty} |f(\xi)| d\xi$$

$$\begin{aligned} \text{Thus, } |\langle u, v \rangle| &= \left| \int_{\mathbb{R}} u(t) v^*(t) dt \right| \\ &\leq \int_{\mathbb{R}} |u(t)| |v(t)| dt \\ &\leq \int_{\mathbb{R}} \frac{1}{2} (|u(t)|^2 + |v(t)|^2) dt \\ &= \frac{1}{2} (\|u\|_2^2 + \|v\|_2^2) \quad \text{--- (A)} \end{aligned}$$

— This can be generalized further.

$$\frac{1}{p} + \frac{1}{q} = 1 \quad 1 < p, q < \infty$$

$$\int_{-\infty}^{+\infty} |u(t)|^p dt < \infty \quad \int_{-\infty}^{+\infty} |v(t)|^q dt < \infty$$

then $t \mapsto u(t)v(t)$ is integrable (due to Hölder's Inequality [Theorem 3.3.2 ahead]).

- The bound (A) can be sharpened further using Cauchy-Schwarz. (As we do now).

SECTION 3.3 The Cauchy-Schwarz Inequality

- One of the most important inequalities on the inner product.

THEOREM 3.3.1 (Cauchy-Schwarz)

$\underline{u}, \underline{v} : \mathbb{R} \rightarrow \mathbb{C}, E_{\underline{u}}, E_{\underline{v}} < \infty$ then $t \mapsto u(t)v^*(t)$ is integrable. and

$$|\langle \underline{u}, \underline{v} \rangle| \leq \|\underline{u}\|_2 \|\underline{v}\|_2$$

OR

$$\left| \int_{-a}^{+a} u(t)v^*(t) dt \right| \leq \sqrt{\int_{-a}^{+a} |u(t)|^2 dt} \sqrt{\int_{-a}^{+a} |v(t)|^2 dt}$$

Equality occurs if $u(t) = \alpha v(t)$.
and
only
if

Proof: $\alpha, \beta \in \mathbb{C}$.

$$\begin{aligned} |\langle \alpha \underline{u}, \beta \underline{v} \rangle| &= |\alpha \langle \underline{u}, \underline{v} \rangle \beta^*| \\ &= |\alpha| |\beta| |\langle \underline{u}, \underline{v} \rangle| \end{aligned}$$

$$\text{But, } |\langle \alpha \underline{u}, \beta \underline{v} \rangle| \leq \frac{1}{2} |\alpha|^2 \|\underline{u}\|_2^2 + \frac{1}{2} |\beta|^2 \|\underline{v}\|_2^2$$

(3.16) or (A)

$$\text{Put } \alpha = \frac{1}{\|\underline{u}\|_2} \quad \beta = \frac{1}{\|\underline{v}\|_2}$$

$$\Rightarrow \frac{1}{\|\underline{u}\|_2} \frac{1}{\|\underline{v}\|_2} |\langle \underline{u}, \underline{v} \rangle| \leq \frac{1}{2} + \frac{1}{2}$$

$$= 1$$

$$\Rightarrow |\langle \underline{u}, \underline{v} \rangle| \leq \|\underline{u}\|_2 \|\underline{v}\|_2.$$

Also, if $\|\underline{u}\|_2 = 0$ or $\|\underline{v}\|_2 = 0 \Rightarrow \langle \underline{u}, \underline{v} \rangle = 0$.

take $|\alpha| = 1$ or $\alpha = 1$.

$$|\beta| |\langle \underline{u}, \underline{v} \rangle| \leq \frac{1}{2} |\beta|^2 \|\underline{v}\|_2^2$$

$$|\langle \underline{u}, \underline{v} \rangle| \leq \frac{1}{2} |\beta| \|\underline{v}\|_2^2$$

$$\beta \rightarrow 0 \Rightarrow |\langle \underline{u}, \underline{v} \rangle| = 0.$$

THEOREM 3.3.2. (Hölder's Inequality)

25

$$|\langle \underline{u}, \underline{v} \rangle| \leq \|\underline{u}\|_p \|\underline{v}\|_q.$$

Assuming $\|\underline{u}\|_p < \infty$, $\|\underline{v}\|_q < \infty$, $1 < p, q < \infty$

$\frac{1}{p} + \frac{1}{q} = 1 \Rightarrow t \mapsto u(t) v^*(t)$ is integrable.

$$\left| \int_{-\infty}^{+\infty} u(t) v^*(t) dt \right| \leq \left(\int_{-\infty}^{+\infty} |u(t)|^p dt \right)^{1/p} \times \left(\int_{-\infty}^{+\infty} |v(t)|^q dt \right)^{1/q}.$$

Proof: see Rudin / Royden.

3.4 Applications of Cauchy-Schwarz.

There are many applications of the Cauchy-Schwarz inequality

We will state and prove a few.

PROPOSITION 3.4.1 (Triangle Inequality for L_2) If $\underline{u}$ and $\underline{v}$ are in L_2 , then:

$$\|\underline{u} + \underline{v}\|_2 \leq \|\underline{u}\|_2 + \|\underline{v}\|_2$$

$$\|\underline{u} + \underline{v}\|_2^2 = \langle \underline{u} + \underline{v}, \underline{u} + \underline{v} \rangle$$

$$= \langle \underline{u}, \underline{u} \rangle + \langle \underline{u}, \underline{v} \rangle + \langle \underline{v}, \underline{u} \rangle + \langle \underline{v}, \underline{v} \rangle$$

$$= \|u\|_2^2 + \langle u, v \rangle + \langle u, v \rangle^* + \|v\|_2^2$$

$$= \|u\|_2^2 + 2\operatorname{Re}\{\langle u, v \rangle\} + \|v\|_2^2$$

$$\leq \|u\|_2^2 + 2|\langle u, v \rangle| + \|v\|_2^2$$

$$\leq \|u\|_2^2 + 2\|u\|_2\|v\|_2 + \|v\|_2^2$$

$$= (\|u\|_2 + \|v\|_2)^2$$

Removing the squares gives:

$$\|u + v\|_2 \leq \|u\|_2 + \|v\|_2$$

Continuity of the Inner product:

A function $f(u, v)$ is continuous at u_0, v_0 if

$$\lim_{n \rightarrow \infty} u_n = u_0$$

$$\lim_{n \rightarrow \infty} v_n = v_0$$

$\Downarrow$

$$\lim_{n \rightarrow \infty} f(u_n, v_n) = f(u_0, v_0).$$

The inner product is a continuous function:

$$\text{meaning } \underline{u}_n \rightarrow \underline{u} \wedge \underline{v}_n \rightarrow \underline{v} \Rightarrow \langle \underline{u}_n, \underline{v}_n \rangle \rightarrow \langle \underline{u}, \underline{v} \rangle$$

$$\text{or } |\langle \underline{u}_n, \underline{v}_n \rangle - \langle \underline{u}, \underline{v} \rangle| \rightarrow 0.$$

$$|\langle \underline{u}_n, \underline{v}_n \rangle - \langle \underline{u}, \underline{v} \rangle|$$

$$= |\langle \underline{u}_n - \underline{u}, \underline{v}_n \rangle + \langle \underline{u}_n - \underline{u}, \underline{v} \rangle|$$

$$= |\langle \underline{u}_n - \underline{u}, \underline{v}_n - \underline{v} \rangle|$$

$$= |\langle \underline{u}_n, \underline{v}_n \rangle - \langle \underline{u}, \underline{v}_n \rangle + \langle \underline{u}, \underline{v}_n \rangle - \langle \underline{u}, \underline{v} \rangle|$$

$$= |\langle \underline{u}_n - \underline{u}, \underline{v}_n \rangle + \langle \underline{u}, \underline{v}_n - \underline{v} \rangle|$$

$$\leq |\langle \underline{u}_n - \underline{u}, \underline{v}_n \rangle| + |\langle \underline{u}, \underline{v}_n - \underline{v} \rangle|$$

$$\leq \|\underline{u}_n - \underline{u}\|_2 \|\underline{v}\|_2 + \|\underline{u}\|_2 \|\underline{v}_n - \underline{v}\|_2$$

$$\downarrow$$

0

$$\downarrow$$

0

$$\rightarrow 0.$$

Another useful consequence of the Cauchy-Schwarz

(28)

Inequality is that if a signal is finite Energy and zero outside a finite Interval then its integrable.

(i) $a, b \in \mathbb{R}, a \leq b.$

(ii) $\int_a^b |x(\xi)|^2 d\xi < \infty$

$$\Rightarrow \int_a^b |x(\xi)| d\xi \leq \sqrt{b-a} \sqrt{\int_a^b |x(\xi)|^2 d\xi}$$

In particular: $\int_a^b |x(\xi)| d\xi < \infty$

PROPOSITION 3.4.3 (Finite-Energy signals over Finite Intervals are Integrable).

Solution (PROOF):

$$\begin{aligned} & \int_{-\infty}^{+\infty} \int_a^b |x(\xi)|^2 d\xi \\ &= \int_{-\infty}^{+\infty} \mathbb{I}\{a \leq \xi \leq b\} |x(\xi)|^2 d\xi \\ &\leq \sqrt{\int_{-\infty}^{+\infty} (\mathbb{I}\{a \leq \xi \leq b\})^2 d\xi} \sqrt{\int_{-\infty}^{+\infty} |x(\xi)|^2 d\xi} \end{aligned}$$

$$= \sqrt{\int_a^b d\xi} \sqrt{\int_{-\infty}^{+\infty} |x(\xi)|^2 d\xi}$$

$$= \sqrt{\xi \Big|_a^b} \sqrt{\int_a^b |x(\xi)|^2 d\xi}$$

$$= \sqrt{b-a} \sqrt{\int_a^b |x(\xi)|^2 d\xi}.$$

Note in general an energy limited signal need not be integrable, e.g.

$$u: t \mapsto \begin{cases} 0 & t \leq 1 \\ 1/t & \text{else} \end{cases}$$

is not integrable but $E_u < \infty$.

— The Cauchy-Schwarz Inequality demonstrates that if

u and v both are $E < \infty \Rightarrow \langle u, v \rangle$ is well defined.

— There can be exceptions: $E_u \neq \infty$ but v decays sufficiently fast for $\langle u, v \rangle$ to exist.

PROPOSITION 3.4.4 If the Lebesgue measurable functions

$x: \mathbb{R} \rightarrow \mathbb{C}$ and $y: \mathbb{R} \rightarrow \mathbb{C}$ satisfy

$$\int_{-\infty}^{+\infty} \frac{|x(t)|^2}{1+t^2} dt < \infty$$

$$\int_{-\infty}^{+\infty} \frac{|y(t)|^2}{1+t^2} dt < \infty$$

$$\int_{-\infty}^{+\infty} |y(t)|^2 (1+t^2) dt < \infty.$$

then $t \mapsto x(t)y^*(t)$ is integrable and:

$$\left| \int_{-\infty}^{+\infty} x(t)y^*(t) dt \right| \leq \sqrt{\int_{-\infty}^{+\infty} \frac{|x(t)|^2}{1+t^2} dt} \sqrt{\int_{-\infty}^{+\infty} |y(t)|^2 (1+t^2) dt}$$

Proof. just use $u(t) = \frac{x(t)}{\sqrt{1+t^2}}$, $v(t) = y(t)\sqrt{1+t^2}$

in the Cauchy-Schwarz Inequality.

3.5: The Cauchy-Schwarz Inequality for random variables.

$$\begin{aligned}
 \text{Recall : } \text{Var}(\underline{X}) &= \overline{(\underline{X} - \overline{\underline{X}})^2} \\
 &= \overline{E[\underline{X} - E[\underline{X}]]^2} \\
 &= E[\underline{X}^2] - (E[\underline{X}])^2
 \end{aligned}$$

(3)

Cauchy-Schwarz works for random Variables by replacing the Integral operator with the expectation operator.

THEOREM 3.5-1 (Cauchy-Schwarz Inequality for Random Variables).

$$|E[UV]| \leq \sqrt{E[U^2]} \sqrt{E[V^2]}$$

With equality iff $\Pr[U = \alpha V] = 1$ $\alpha \neq 0$.

$$\begin{aligned}
 E[UV] &= \int_{-\infty}^{+\infty} E[UV | V=v] P_V(v) dv \\
 &= \int_{-\infty}^{+\infty} E[Uv | V=v] P_V(v) dv \\
 &= \int_{-\infty}^{+\infty} v E[U | V=v] P_V(v) dv \\
 &= \int_{-\infty}^{+\infty} v \int_{-\infty}^{+\infty} u P_{U|V}(u|v) du P_V(v) dv \\
 &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} vu P_{U,V}(u,v) du dv
 \end{aligned}$$

$$= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} uv \sqrt{p_{\underline{u}, \underline{v}}(u, v)} \sqrt{p_{\underline{u}, \underline{v}}(u, v)} du dv$$

$$= \sqrt{\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} u^2 p_{\underline{u}, \underline{v}}(u, v) du dv} \sqrt{\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} v^2 p_{\underline{u}, \underline{v}}(u, v) du dv}$$

$$= \sqrt{\int_{-\infty}^{+\infty} u^2 \int_{-\infty}^{+\infty} p_{\underline{u}, \underline{v}}(u, v) dv du} \sqrt{\int_{-\infty}^{+\infty} v^2 \int_{-\infty}^{+\infty} p_{\underline{u}, \underline{v}}(u, v) du dv}$$

$$= \sqrt{\int_{-\infty}^{+\infty} u^2 p(u) du} \sqrt{\int_{-\infty}^{+\infty} v^2 p(v) dv}$$

$$= \sqrt{E[u^2]} \sqrt{E[v^2]} \quad \square.$$

For the next Corollary recall:

$$\text{Cov}(u, v) = E[(u - E[u])(v - E[v])].$$

COROLLARY 3.52. (Covariance Inequality) If the R.V.s $\underline{u}, \underline{v}$

are of finite variance, $\text{Var}[u]$ and $\text{Var}[v]$. then

→

$$|\text{Cov}[u, v]| \leq \sqrt{\text{Var}[u]} \sqrt{\text{Var}[v]}$$

Proof:

We have

$$\begin{aligned}
 |\text{Cov}[u, v]| &= |E[(u - E[u])(v - E[v])]| \\
 &\leq \sqrt{E[(u - E[u])^2]} \sqrt{E[(v - E[v])^2]} \\
 &= \sqrt{\text{Var}[u]} \sqrt{\text{Var}[v]}
 \end{aligned}$$

The correlation coefficient is defined as.

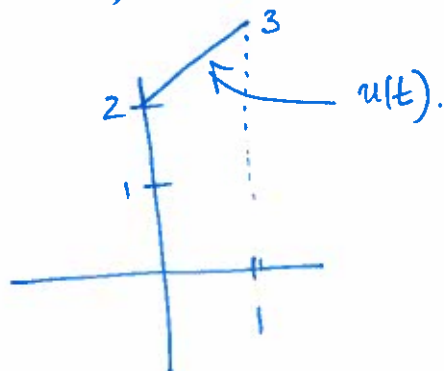
$$\rho = \frac{\text{Cov}[u, v]}{\sqrt{\text{Var}[u]} \sqrt{\text{Var}[v]}}$$

$$-1 \leq \rho \leq +1$$

EXERCISES

EXERCISE 5.1 (Truncated Polynomials) Consider a signal

$$u: t \mapsto (t+2) \mathbb{I}_{\{0 \leq t \leq 1\}}$$



and $\underline{v}: t \mapsto (t^2 - 2t - 3) \mathbb{I} \{0 \leq t \leq 1\}$. Compute the energies (34)

$\|\underline{u}\|_2^2$ and $\|\underline{v}\|_2^2$ as well as the inner product $\langle \underline{u}, \underline{v} \rangle$.

$$\begin{aligned} \|\underline{u}\|_2^2 &= \langle \underline{u}, \underline{u} \rangle = \int_0^1 (t+2)^2 dt \\ &= \left. \frac{(t+2)^3}{3} \right|_0^1 \\ &= \frac{3^3}{3} - \frac{2^3}{3} \\ &= \frac{27-8}{3} = \frac{19}{3}. \end{aligned}$$

⚡ This should be -, correct all that follows.

$$\|\underline{v}\|_2^2 = \int_0^1 (t^2 - 2t + 3)^2 dt$$

$$= \int_0^1 t^4 + 4t^2 + 9 - 2t^3 + 3t^2 \cdot 2 - 4 \cdot 3t dt$$

$$= \left. \frac{t^5}{5} \right|_0^1 + \left. \frac{4t^3}{3} \right|_0^1 + 9t \Big|_0^1 - \left. \frac{4t^4}{4} \right|_0^1 + \left. \frac{6 \cdot t^3}{3} \right|_0^1 - \left. \frac{12t^2}{2} \right|_0^1$$

$$= \frac{1}{5} + \frac{4}{3} + 9 - 1 + 2 - 6$$

$$= \frac{1}{5} + \frac{4}{3} + 4 = \frac{3+20+15 \cdot 4}{15}$$

$$= \frac{3+20+60}{15}$$

$$= \frac{83}{15}.$$

$$\langle \underline{u}, \underline{v} \rangle = \int_0^1 (t+2)(t^2-2t-3) dt$$

$$= \int_0^1 (t^3 - 2t^2 - 3t + 2t^2 - 4t - 6) dt$$

$$= \int_0^1 (t^3 - 7t - 6) dt$$

$$= \left. \frac{t^4}{4} \right|_0^1 - \left. \frac{7t^2}{2} \right|_0^1 - 6t \Big|_0^1$$

$$= \frac{1}{4} - \frac{7}{2} - 6$$

$$= \frac{2 - 28 - 48}{8}$$

$$= -\frac{74}{8} \text{ Ans.}$$

EXERCISE 3.2 (Inner products of mirror images)

Express $\langle \hat{x}, \hat{y} \rangle$ in terms of $\langle x, y \rangle$

$$\langle \hat{x}, \hat{y} \rangle = \int_{-\infty}^{+\infty} x(-t) y(-t) dt$$

$$= \int_{+\infty}^{-\infty} x(u) y(u) du$$

$$\begin{aligned} -t &= u \\ t &= -u \\ dt &= -du. \end{aligned}$$

$$= \int_{-\infty}^{+\infty} x(u) y(u) du.$$

EXERCISE 3.3 (Finite Energy Signals) Let $\underline{x}$ be an energy-limited signal.

(3c)

(i) show that for all $t \in \mathbb{R}$ the signal $t \mapsto x(t-t)$ must also be energy-limited. (Obvious)

(ii) Show that the reflection of the signal x must also be energy limited (Obvious) $[t \mapsto x(-t)]$

(iii) How are the energies of $t \mapsto x(t)$, $t \mapsto x(t-t)$ and $t \mapsto x(-t)$ related. (Obvious: they are equal).

EXERCISE 3.4: Show that if $\underline{u}$, $\underline{v}$ and $\underline{w}$ are energy-limited signals then

$$\begin{aligned} \langle \underline{u} + \underline{v} + 3\underline{u} + \underline{v} + i\underline{w} \rangle &= 3\|\underline{u}\|_2^2 + \|\underline{v}\|_2^2 + \langle \underline{u}, \underline{v} \rangle \\ &\quad + 3\langle \underline{u}, \underline{v} \rangle^* - i\langle \underline{u}, \underline{w} \rangle - i\langle \underline{v}, \underline{w} \rangle \end{aligned}$$



$$\langle \underline{u}, 3\underline{u} + \underline{v} + i\underline{w} \rangle + \langle \underline{v}, 3\underline{u} + \underline{v} + i\underline{w} \rangle$$

$$\langle \underline{u}, 3\underline{u} \rangle + \langle \underline{u}, \underline{v} \rangle + \langle \underline{u}, i\underline{w} \rangle + 3\langle \underline{v}, \underline{u} \rangle + \langle \underline{v}, \underline{v} \rangle +$$

$$\langle \underline{v}, i\underline{w} \rangle$$

$$= 3\|\underline{u}\|_2^2 + \|\underline{v}\|_2^2 + \langle \underline{u}, \underline{v} \rangle + 3\langle \underline{u}, \underline{v} \rangle^* - i\langle \underline{u}, \underline{w} \rangle$$

$$- \langle \underline{v}, \underline{w} \rangle \quad \square.$$

(3)

EXERCISE 3.5: (A useful identity)

$$a^2 - b^2 = (a-b)(a+b) \quad (\text{Classic algebra})$$

Similarly show that,

$$\|\underline{u}\|_2^2 - \|\underline{v}\|_2^2 = \langle \underline{u} - \underline{v}, \underline{u} + \underline{v} \rangle$$

$$\langle \underline{u} - \underline{v}, \underline{u} + \underline{v} \rangle = \langle \underline{u}, \underline{u} + \underline{v} \rangle - \langle \underline{v}, \underline{u} + \underline{v} \rangle$$

$$= \langle \underline{u}, \underline{u} \rangle + \langle \underline{u}, \underline{v} \rangle - \langle \underline{v}, \underline{u} \rangle + \langle \underline{v}, \underline{v} \rangle$$

$$= \|\underline{u}\|_2^2 - \|\underline{v}\|_2^2.$$

(b) Does it also hold for complex signals?

No. Check it using $\underline{u} = a + bi$, $\underline{v} = c + di$

EXERCISE 3.6: (Orthogonality to all signals) Let $\underline{u}$ be an energy limited signal to show that,

$$(\underline{u} \equiv 0) \Leftrightarrow (\langle \underline{u}, \underline{v} \rangle = 0, \forall \underline{v} \in \mathcal{L}_2)$$

we use the following fact:

(38)

(PROPOSITION 2.5.3) (page 9)

(i) A non-negative measurable signal integrates to zero if and only if, it is indistinguishable for 0 signal. i.e., if $\mu(\{t: x(t) \neq 0\}) = 0$

$\uparrow$ This is the definition of $\underline{0} \equiv \underline{x}$.

Let $u(t)$ be integrable. $\Rightarrow \langle u, 1 \rangle$ is defined.

$$\int |u(t)| \cdot 1 dt = 0 \Leftrightarrow \langle u, 1 \rangle = 0$$

$$\Updownarrow$$
$$u \equiv 0 \quad (\text{by Proposition 2.5.3 above})$$

If $\|u\|_2^2 < \infty$ i.e. u is an energy limited function.

then we need a different strategy.

$$\text{Suppose } \underline{u} \equiv 0 \Rightarrow \mu(\{|u(t)|^2 \neq 0\}) = 0.$$

$$\text{because } \underbrace{\{|u(t)|^2 \neq 0\}}_A \Rightarrow \underbrace{\{|u(t)| \neq 0\}}_B$$

$$t \in A \Rightarrow t \in B$$

$$A \subseteq B.$$

Thus, $\mu(\{|u(t)|^2 \neq 0\}) = 0.$

$$\text{and } |u(t)|^2 \equiv 0$$

$$\Downarrow$$

$$\int_{\mathbb{R}} |u(t)|^2 dt = 0$$

$$\Downarrow$$

$$\|u\|_2^2 = 0 \Rightarrow \|u\|_2 = 0$$

$$\Downarrow$$

$$|\langle u, v \rangle| \leq \|u\|_2 \|v\|_2 = 0$$

$$\Downarrow$$

$$\langle u, v \rangle = 0 \quad \forall v$$

Also if $\langle u, v \rangle = 0 \quad \forall v \Rightarrow \langle u, u \rangle = 0 \Rightarrow \|u\|_2^2 = 0$

$$\Rightarrow \int |u(t)|^2 dt = 0 \Rightarrow \int |u(t)| dt = 0 \Rightarrow u \equiv 0.$$

□.

EXERCISE 3.7 (On the tightness of the Cauchy-Schwarz

inequality). Show that the bound obtained

from the Cauchy-Schwarz inequality is at least

as (3.16) $\rightarrow |\langle u, v \rangle| = \frac{1}{2} (\|u\|_2^2 + \|v\|_2^2)$

$$|\langle u, v \rangle| \leq \|u\|_2 \|v\|_2$$

We have $\|u\|_2 \|v\|_2 \leq \frac{1}{2} (\|u\|_2^2 + \|v\|_2^2)$

$$\begin{aligned} 0 &\leq \|u\|_2^2 + \|v\|_2^2 - 2\|u\|_2 \|v\|_2 \\ &= (\|u\|_2 - \|v\|_2)^2 \quad \square. \end{aligned}$$

EXERCISE 3.8 (A Generalized Cauchy-Schwarz inequality)

Let $h: \mathbb{R} \rightarrow \mathbb{R}$ be a non-negative function. Prove if the complex signals u and v are such that the integrals $\int |u(t)|^2 h(t) dt$ and $\int |v(t)|^2 h(t) dt$ are finite then:

$$\left| \int_{-\infty}^{+\infty} u(t) v^*(t) h(t) dt \right|^2 \leq \int_{-\infty}^{+\infty} |u(t)|^2 h(t) dt \int_{-\infty}^{+\infty} |v(t)|^2 h(t) dt.$$

This is easy, just take $u_1 = u(t)\sqrt{h(t)}$
Solution. $u_2 = v(t)\sqrt{h(t)}$

EXERCISE 3.9: (Indistinguishability and Inner Product)

Let $u \in \mathcal{L}_2$ be indistinguishable from $u' \in \mathcal{L}_2$ and $v \equiv v' \in \mathcal{L}_2$.

Show that $\langle u, v \rangle = \langle u', v' \rangle$.

$$\begin{aligned}
 & |\langle u, v \rangle - \langle u', v' \rangle| \\
 &= |\langle u, v \rangle - \langle u', v \rangle + \langle u', v \rangle + \langle u', v' \rangle| \\
 &= |\langle u - u', v \rangle + \langle u', v - v' \rangle| \\
 &\leq |\langle u - u', v \rangle| + |\langle u', v - v' \rangle| \\
 &\leq \underbrace{\|u - u'\|_2}_{\downarrow 0} \|v\|_2 + \|u'\|_2 \underbrace{\|v - v'\|_2}_{\downarrow 0}.
 \end{aligned}$$

because, $u \equiv u'$

$$u - u' \equiv 0$$

$$(u - u')^2 \equiv 0.$$

$$\|u - u'\|_2 = 0.$$

EXERCISE 3.10 (Finite Energy and Integrability)

42

Let $\underline{x}: \mathbb{R} \rightarrow \mathbb{C}$ be Lebesgue measurable.

(i) Show that $t \mapsto \underline{x}(t)$ is $E < \infty$ $\wedge$ $t \mapsto t \underline{x}(t) \stackrel{!}{\neq} E < \infty$



$$\int_{-\infty}^{+\infty} |\underline{x}(t)|^2 (1+t^2) dt < \infty \quad (3.29)$$

(ii) Show that (3.29) implies that $\underline{x}$ is integrable.

(iii) Give an example of an integrable system that does not satisfy (3.29).

(i) is obvious.

(ii) is a bit tricky $\int |\underline{x}(t)| dt = \int |\underline{x}(t)| \frac{1}{\sqrt{1+t^2}} \cdot \sqrt{1+t^2} dt$

$$\leq \left(\int_{-\infty}^{+\infty} \frac{1}{1+t^2} dt \right)^{1/2} \left(\int_{-\infty}^{+\infty} |\underline{x}(t)|^2 (1+t^2) dt \right)^{1/2}$$

$$\leq \left[\operatorname{Arctan} \Big|_{-\infty}^{+\infty} \right]^{1/2} \cdot K$$

$$= \left(\frac{\pi}{2} - \left(-\frac{\pi}{2} \right) \right)^{1/2} K$$

$$= \sqrt{\pi} K < \infty.$$

(iii) we seek a function for which $x \in L_1$ but

(43)

$$\int |x(t)|^2 (1+t^2) dt < \infty \text{ or } x \notin L_2.$$

$$\text{take } x(t) = \begin{cases} \frac{1}{t^{2/3}} & 0 < t \leq 1 \\ 0 & t=0, \text{ else} \end{cases}$$

$$\int_0^1 |x(t)| dt = \int_0^1 \frac{1}{t^{1/3}} dt = 3t^{2/3} \Big|_0^1 = 3 < \infty$$

$$\begin{aligned} \text{However } \int |x(t)|^2 dt &= \int_0^1 t^{-4/3} dt \\ &= -3t^{-1/3} \Big|_0^1 \\ &= 3t^{-1/3} \Big|_1^0 \\ &= \infty. \end{aligned}$$

Thus $x(t) \in L_1$ but $x(t) \notin L_2$.

□

EXERCISE 3.11 (The Variance of a Sum) Show that if

random variables $\underline{X}$ and $\underline{Y}$ are of finite variance,

$$\text{then: } \text{Var}[\underline{X} + \underline{Y}] \leq \left(\sqrt{\text{Var}(\underline{X})} + \sqrt{\text{Var}(\underline{Y})} \right)^2$$

$$\text{Var}[\underline{X} + \underline{Y}] = E[\underline{X} + \underline{Y} - E[\underline{X} + \underline{Y}]]^2$$

(44)

$$= E\left[\left((\underline{X} - E[\underline{X}]) + (\underline{Y} - E[\underline{Y}])\right)^2\right]$$

$$= \text{Var}(\underline{X}) + \text{Var}(\underline{Y}) + 2E[(\underline{X} - E[\underline{X}])(\underline{Y} - E[\underline{Y}])]$$

$$= \text{Var}(\underline{X}) + \text{Var}(\underline{Y}) + 2\text{Cov}(\underline{X}, \underline{Y})$$

$$\leq \text{Var}(\underline{X}) + \text{Var}(\underline{Y}) + 2\sqrt{\text{Var}(\underline{X})}\sqrt{\text{Var}(\underline{Y})}$$

$$= \left(\sqrt{\text{Var}(\underline{X})} + \sqrt{\text{Var}(\underline{Y})}\right)^2.$$

EXERCISE 3.12 The Cauchy-Schwarz Inequality for Sequences.

(i) Let $a_1, a_2, \dots$; $b_1, b_2, \dots$ satisfy:

$$\sum_i |a_i|^2 < \infty ; \sum_i |b_i|^2 < \infty$$

Show that:

$$\left| \sum_i a_i b_i^* \right|^2 \leq \left(\sum_i |a_i|^2 \right) \left(\sum_i |b_i|^2 \right)$$

The proof is quite similar for that for integrals. (45)

$$\begin{aligned} |\langle u, v \rangle| &= \left| \sum_{i=1}^{\infty} a_i b_i^* \right| \\ &\leq \sum_{i=1}^{\infty} |a_i| |b_i| \\ &\leq \sum_{i=1}^{\infty} \frac{1}{2} (|a_i|^2 + |b_i|^2) \\ &= \frac{1}{2} \left(\sum_{i=1}^{\infty} |a_i|^2 + \sum_{i=1}^{\infty} |b_i|^2 \right) \\ &= \frac{1}{2} (\|u\|_2^2 + \|v\|_2^2) \end{aligned}$$

Thus $|\langle \alpha u, \beta v \rangle| = |\alpha| |\beta| |\langle u, v \rangle|$

$$\leq \frac{1}{2} (|\alpha|^2 \|u\|_2^2 + |\beta|^2 \|v\|_2^2)$$

Put $|\alpha| = \frac{1}{\|u\|}$, $|\beta| = \frac{1}{\|v\|}$

$$= \frac{1}{2} (1 + 1) = 1$$

$$\frac{1}{\|u\|_2 \|v\|_2} |\langle u, v \rangle| \leq 1$$

$$|\langle u, v \rangle| \leq \|u\|_2 \|v\|_2$$

(ii) Derive Cauchy-Schwarz for d-tuples:

(46)

$$\left| \sum_{i=1}^d a_i b_i^* \right|^2 \leq \left(\sum_{i=1}^d a_i^2 \right) \left(\sum_{i=1}^d b_i^2 \right)$$

This is easy, just define $\{a'_i\}_{i=1}^\infty, \{b'_i\}_{i=1}^\infty$ as.

$$a'_i = \begin{cases} a_i & 1 \leq i \leq d \\ 0 & \text{else} \end{cases}$$

$$b'_i = \begin{cases} b_i & 1 \leq i \leq d \\ 0 & \text{else} \end{cases}$$

and apply the standard Cauchy-Schwarz, for infinite sequences.

EXERCISE 3.13 (Cauchy-Schwarz and Matrices)

$$A, B \in \mathbb{R}^{n \times m} \Rightarrow$$

$$\begin{aligned} \left(\text{tr}(AB^T) \right)^2 &\leq \left(\sum_{k=1}^n \sum_{\ell=1}^m (a^{(k,\ell)})^2 \right) \\ &\quad \times \left(\sum_{k=1}^n \sum_{\ell=1}^m (b^{(k,\ell)})^2 \right) \end{aligned}$$

(Hint: $\mapsto$ Ex 3.12 (ii).)

This is a trivial application of Cauchy-Schwarz for (47)

finite sequences.

$$\text{tr}(AB^T) = \sum_{k=1}^n \left(\sum_{\ell=1}^m a_{k\ell} b_{k\ell} \right) \quad B_{\ell,k}^T = b_{k\ell}$$

Thus applying Cauchy-Schwarz to the double summation:

$$\left(\sum_{k=1}^n \sum_{\ell=1}^m a_{k\ell} b_{k\ell} \right)^2 \leq \left(\sum_{k=1}^n \sum_{\ell=1}^m (a_{k\ell})^2 \right) \left(\sum_{k=1}^n \sum_{\ell=1}^m (b_{k\ell})^2 \right)$$

EXERCISE 3.14. (Summability and Square Summability)

Let, $a_1, a_2, \dots$ be a sequence of complex numbers.

Show that:

$$\sum_{i=1}^{\infty} |a_i| < \infty \Rightarrow \left(\sum_{i=1}^{\infty} |a_i|^2 < \infty \right)$$

Solution: This is a bit tricky:

$$\begin{aligned} \sum_i |a_i|^2 &= \sum_i |a_i| |a_i| \\ &= \sum_i \frac{|a_i| |a_i| \sqrt{(1+|a_i|)}}{\sqrt{(1+|a_i|)}} \\ &\leq \left[\sum_i \frac{|a_i|^2}{(1+|a_i|)} \right]^{1/2} \left[\sum_i |a_i|^2 (1+|a_i|) \right]^{1/2} \end{aligned}$$

Now $\sum |a_i| < \infty \Rightarrow \lim_i |a_i| = 0 \Rightarrow \exists k$ such that
 $|a_i| < k \quad \forall i \in \mathbb{N}$.

$$\begin{aligned} \text{Thus,} \quad &= \left(\sum_{i=1}^{\infty} |a_i| \frac{|a_i|}{1+|a_i|} \right)^{1/2} \left(\sum_{i=1}^{\infty} |a_i|^2 (1+|a_i|) \right)^{1/2} \\ &\leq \left(\sum_i |a_i| \right)^{1/2} \left(\sum |a_i|^2 (1+k) \right)^{1/2} \end{aligned}$$

$$\text{Thus,} \quad \left(\sum_{i=1}^{\infty} |a_i|^2 \right)^2 \leq \sum_{i=1}^{\infty} |a_i| \cdot (1+k) \sum_{i=1}^{\infty} |a_i|^2$$

$$\sum_{i=1}^{\infty} |a_i| \leq \sum_{i=1}^{\infty} |a_i| (1+k) < \infty. \quad \square$$

EXERCISE 3.15. (A Friendlier GPA) Use Cauchy-Schwarz
 for d -tuples to show that:

$$\frac{a_1 + a_2 + \dots + a_d}{d} \leq \sqrt{\frac{a_1^2 + a_2^2 + \dots + a_d^2}{d}}$$

$$a_1, a_2, \dots, a_d \in \mathbb{R}.$$

Solution: Straight forward application of C.S.

$$\frac{a_1 + a_2 + \dots + a_d}{d} = \sum a_i \cdot \frac{1}{d}$$

$$\leq \sqrt{\sum_i^d a_i^2} \sqrt{\sum_i^d \frac{1}{d^2}}$$

$$= \frac{\sqrt{a_1^2 + a_2^2 + \dots + a_d^2}}{\sqrt{d/d^2}}$$

$$= \sqrt{\frac{a_1^2 + a_2^2 + \dots + a_d^2}{d}} \quad \square$$

Chapter 4: The space L_2 of Energy-Limited Signals.

26 July 2025

(50)

- The space L_2 is a vector space.

- L_2 was defined as:

$$\underline{u} : \int_{\mathbb{R}} |u(t)|^2 dt < \infty$$

- The "length" or norm on the signal $\underline{u}$ is

$$\|\underline{u}\|_2 = \sqrt{\int_{-\infty}^{+\infty} |u(t)|^2 dt}$$

- We denote $\underline{u}$ with $E_u < \infty$.

SECTION 4.2: L_2 as a vector space.

Define, $\underline{u} + \underline{v} : t \mapsto u(t) + v(t)$

$\alpha \in \mathbb{C}$, $\alpha \underline{u} : t \mapsto \alpha u(t)$.

To show that L_2 is a vector space

we need to show that $\underline{u} \in L_2$, $\underline{v} \in L_2$

$\Rightarrow \alpha \underline{u} + \beta \underline{v} \in L_2$.

$$\textcircled{1} \quad \alpha \in \mathbb{C}, \quad t \mapsto \alpha u(t)$$

(5)

$$\begin{aligned} & \int_{\mathbb{R}} |\alpha u(t)|^2 dt \\ &= |\alpha|^2 \int_{\mathbb{R}} |u(t)|^2 dt < \infty. \quad (u \in L_2) \end{aligned}$$

Thus, $\alpha u \in L_2$.

$$\textcircled{2} \quad u, v \in L_2. \quad \|u + v\|_2 \leq \|u\|_2 + \|v\|_2$$

$$\Rightarrow \left(\int_{\mathbb{R}} |u(t) + v(t)|^2 dt \right)^{1/2} \leq \left(\int_{\mathbb{R}} |u(t)|^2 dt \right)^{1/2} + \left(\int_{\mathbb{R}} |v(t)|^2 dt \right)^{1/2}$$

$$\begin{aligned} \int_{\mathbb{R}} |u(t) + v(t)|^2 dt &\leq \int_{\mathbb{R}} |u(t)|^2 dt \\ &\quad + \int_{\mathbb{R}} |v(t)|^2 dt \\ &\quad + 2 \sqrt{\int_{\mathbb{R}} |u(t)|^2 dt \times \int_{\mathbb{R}} |v(t)|^2 dt} \\ &< \infty. \end{aligned}$$

Thus, $u + v \in L_2$

and L_2 is a vector space.

As a vector space the following are obvious:

(53)

(i) Commutativity: $\underline{u} + \underline{v} = \underline{v} + \underline{u}$, $\underline{u}, \underline{v} \in \mathcal{L}_2$

(ii) Associativity: $\underline{u} + (\underline{v} + \underline{w}) = (\underline{u} + \underline{v}) + \underline{w}$.

$$(\alpha\beta)\underline{u} = \alpha(\beta\underline{u}) \quad \alpha, \beta \in \mathbb{C}, \underline{u} \in \mathcal{L}_2.$$

(iii) Additive identity: The all zero signal $\underline{0}: t \mapsto 0$.

$$\underline{0} + \underline{u} = \underline{u} = \underline{u} + \underline{0}, \quad \underline{u} \in \mathcal{L}_2.$$

(iv) Additive inverse: $\forall \underline{v}, \exists \underline{w}$ s.t. $\underline{v} + \underline{w} = \underline{w} + \underline{v} = \underline{0}$

(v) Multiplicative Identity: $1\underline{w} = \underline{w}$

(vi) Distributive properties:

$$\alpha(\underline{u} + \underline{v}) = \alpha\underline{u} + \alpha\underline{v}$$

$$(\alpha + \beta)\underline{u} = \alpha\underline{u} + \beta\underline{u}.$$

signals $\equiv$ vectors

addition $\equiv$ superposition.

scalar multiplication $\equiv$ amplification.

SECTION 4.3: Subspaces, Dimension and Basis.

We borrow from the language of linear algebra:

- Subspaces (linear subspace) of L_2 is a non-empty subset $\mathcal{U} \subseteq L_2$ that is ~~also~~ closed under superposition and amplification:

$$\underline{u}_1, \underline{u}_2 \in \mathcal{U} \Rightarrow \underline{u}_1 + \underline{u}_2 \in \mathcal{U}$$

$$\alpha \in \mathbb{C}, \underline{u} \in \mathcal{U} \Rightarrow \alpha \underline{u} \in \mathcal{U}.$$

Example 4.3.1. Consider the sets of functions of the form $t \mapsto p(t)e^{-|t|}$; $\deg\{p(t)\} \leq 3$

Class exercise: Prove that the above set of functions is a subspace of L_2 .

- Inclusion
- ① Closure in L_2
 - ② Amplification
 - ③ Superposition.

Linear Combination: Let $\{\underline{v}_1, \dots, \underline{v}_n\}$ be list of

55

n vectors (signals) and $\{\alpha_1, \dots, \alpha_n\}$ be a list of scalars.

Then,

$$\sum_{i=1}^n \alpha_i \underline{v}_i = \alpha_1 \underline{v}_1 + \alpha_2 \underline{v}_2 + \dots + \alpha_n \underline{v}_n$$

is called a linear combination of the set $\{\underline{v}_1, \dots, \underline{v}_n\}$.

The all zero-signal $\underline{0}$ is a linear combination of any set of $\underline{v}_n$ s.

Span of a set: $\text{span}(\underline{v}_1, \dots, \underline{v}_n) = \left\{ \sum_{i=1}^n \alpha_i \underline{v}_i : \alpha_1, \dots, \alpha_n \in \mathbb{C} \right\}$.

Thus span is the set of linear combinations in which the scalars α_i 's are allowed to take on any value.

① $\text{span}(\underline{v}_1, \dots, \underline{v}_n)$ is a linear subspace of $\mathcal{L}_2$.

② Suppose $\mathcal{U}$ is a linear subspace of $\mathcal{L}_2$.

and $\{\underline{u}_1, \dots, \underline{u}_n\} \subseteq \mathcal{U}$. The $\text{span}(\underline{u}_1, \dots, \underline{u}_n)$

is a linear subspace of $\mathcal{U}$.

Finite and Infinite dimensional subspaces.

If there is a finite set $\{u_1, \dots, u_n\}$ s.t. $\text{span}(u_1, \dots, u_n) = \mathcal{U}$ then $\mathcal{U}$ is called finite dimensional, otherwise its infinite dimensional. Example:

$\{\underline{u} : t \mapsto p(t)e^{-|t|}\}$: $p(t)$ polynomial is infinite dimensional.

$\{\underline{u} : t \mapsto p(t)e^{-|t|}, \text{degree}(p(t)) \leq 5\}$ is finite dimensional.

— $\mathcal{U}$ is finite dim. $\mathcal{U}' \subseteq \mathcal{U} \Rightarrow \dim(\mathcal{U}') < \infty$.

Linear Independence:

$$\left(\sum \alpha_i \underline{v}_i = \underline{0} \right) \Rightarrow \left(\alpha_1 = \alpha_2 = \dots = \alpha_n = 0 \right)$$

Example $v_1 : t \mapsto e^{-|t|}$

$$v_2 : t \mapsto t e^{-|t|}$$

$$v_3 : t \mapsto t^2 e^{-|t|},$$

If $(v_1, \dots, v_n)$ is NOT linearly independent we say that they are linearly dependent.

Example: $v_1: t \mapsto e^{-|t|}$
 $v_2: t \mapsto t e^{-|t|}$
 $v_3: t \mapsto (2t+1) e^{-|t|}$

Class exercise: Check the above two examples.

A BASIS: d -tuple $(u_1, \dots, u_d)$ is called a basis for the linear subspace $\mathcal{U}$ if it is linearly independent and if $\text{span}(u_1, \dots, u_n) = \mathcal{U}$.

— If $(u_1, \dots, u_d)$ is a basis for $\mathcal{U}$ then

$$\forall v \in \mathcal{U} \exists \alpha_1, \alpha_2, \dots, \alpha_d \in \mathbb{C} \text{ s.t.}$$

$$\underline{v} = \alpha_1 \underline{u}_1 + \alpha_2 \underline{u}_2 + \dots + \alpha_d \underline{u}_d \quad (4.10)$$

— The condition that $(u_1, \dots, u_d)$ is linearly

independent implies that if representation (4.16)

(58)

exists it is unique.

— Thus $(u_1, \dots, u_d)$ is a basis for U if $\text{span}(u_1, \dots, u_n)$

$\subseteq U$ and every $v \in U$ has a unique representation

(4.16).

— Every finite dimensional vector space has a basis.

— All bases for a vector space have the same

of elements. — called the DIMENSION of the

vector space. $\left((u_1, \dots, u_d) \text{ is a basis of } U, (u'_1, \dots, u'_{d'}) \right.$
 $\left. \text{is another basis of } U \Rightarrow d = d' \right).$

— $\dim(\{0\}) = 0$.

— $U' \subseteq U \Rightarrow \dim(U') \leq \dim(U)$.

$u(\cdot)$

KEY PROPERTIES

PROPOSITION 4.4.1 (Properties of $\|\cdot\|_2$) Let $\underline{u}, \underline{v} \in \mathcal{L}_2$ and let $\alpha \in \mathbb{C}$. Then.

$$\|\alpha \underline{u}\|_2 = |\alpha| \|\underline{u}\|_2 \quad (4.11)$$

$$\|\underline{u} + \underline{v}\|_2 \leq \|\underline{u}\|_2 + \|\underline{v}\|_2 \quad (4.12)$$

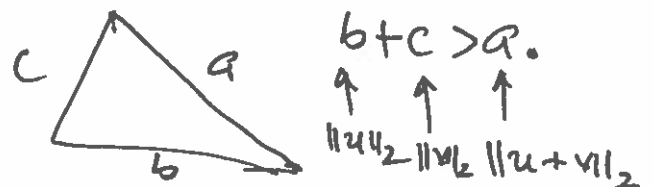
$$(\|\underline{u}\|_2 = 0) \iff (\underline{u} \equiv 0) \quad (4.13)$$

PROOF: (Look it up on perplexity.ai).

(4.11) $\rightarrow$ stretching a vector scales its length.

(4.12) $\rightarrow$ triangle inequality for three sides

a triangle:

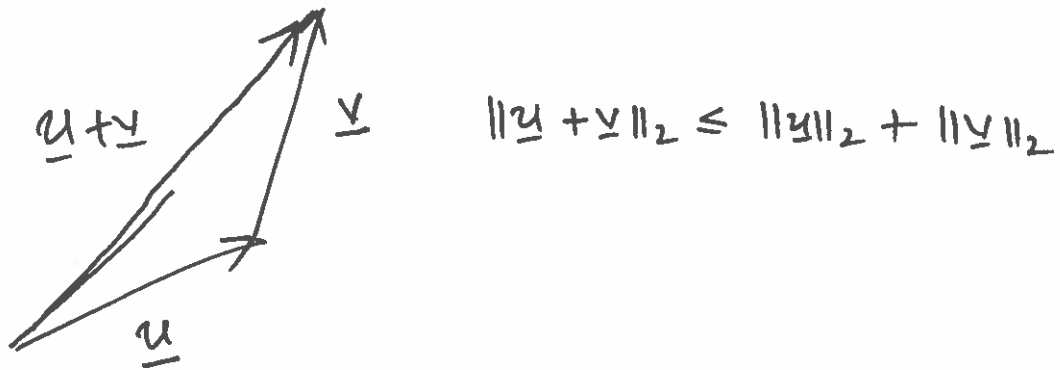


We also have:

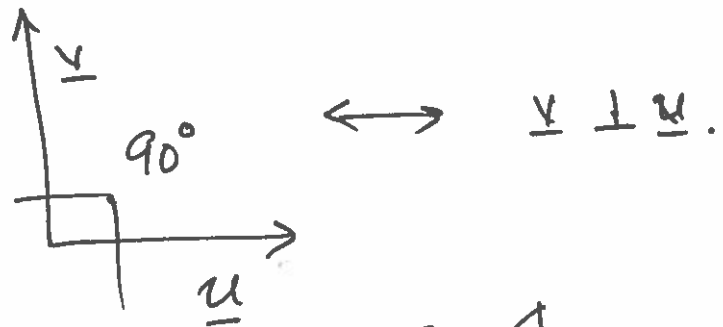
(60)

$$|\|x\|_2 - \|y\|_2| \leq \|x+y\|_2 \leq \|x\|_2 + \|y\|_2.$$

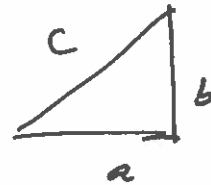
Class work: $\uparrow$ prove it.



SECTION 4.5* ORTHOGONALITY AND INNER PRODUCTS



Idea: Generalize $\underline{a^2 + b^2 = c^2}$



Pythagoras theorem

to

L_2 .

Pythagoras Theorem in L_2 :

61

$$\underline{u} \perp \underline{v} \iff \|\underline{u}\|_2^2 + \|\underline{v}\|_2^2 = \|\underline{u} + \underline{v}\|_2^2$$

$$\begin{cases} \underline{u}: \mathbb{R} \rightarrow \mathbb{C} \\ \underline{v}: \mathbb{R} \rightarrow \mathbb{C} \end{cases}$$

Recall the definition of $\|\cdot\|_2$ (4.2)

$$\begin{aligned} \|\underline{u} + \underline{v}\|_2^2 &= \int_{\mathbb{R}} |u(t) + v(t)|^2 dt \\ &= \int_{\mathbb{R}} (u(t) + v(t))(u(t) + v(t))^* dt \end{aligned}$$

$$\begin{aligned} (w + w^* = 2\operatorname{Re}\{w\}) &= \int_{\mathbb{R}} (|u(t)|^2 + |v(t)|^2 + 2\operatorname{Re}\{u(t)v^*(t)\}) dt \\ &= \|\underline{u}\|_2^2 + \|\underline{v}\|_2^2 + 2\operatorname{Re}\int_{\mathbb{R}} \{u(t)v^*(t)\} dt. \end{aligned}$$

$$\text{So, } \|\underline{u}\|_2^2 + \|\underline{v}\|_2^2 = \|\underline{u} + \underline{v}\|_2^2$$



$$\operatorname{Re}\left\{\int_{\mathbb{R}} u(t)v^*(t) dt\right\} = 0$$

This approach works well for real signals.

$$\underline{u} \perp \underline{v} \iff \langle \underline{u}, \underline{v} \rangle = 0$$

But for complex signals it could be that two signals $\underline{u}, \underline{v}$ are orthogonal but their scaled versions are not. Example: $t \mapsto i \mathbb{I}\{|t| \leq 5\}$ is orthogonal to $t \mapsto \mathbb{I}\{|t| \leq 17\}$ but if you scale it with $\alpha = i$

$$\begin{aligned} \text{then} \quad & \langle i \cdot i \mathbb{I}\{|t| \leq 5\}, \mathbb{I}\{|t| \leq 17\} \rangle \\ &= i^2 \langle \mathbb{I}\{|t| \leq 5\}, \mathbb{I}\{|t| \leq 17\} \rangle \\ &= - \langle \mathbb{I}\{|t| \leq 5\}, \mathbb{I}\{|t| \leq 17\} \rangle \neq 0. \end{aligned}$$

To avoid this "corner case" we define $\underline{u}$ to be orthogonal to $\underline{v}$ if

$$\| \alpha \underline{u} + \underline{v} \|_2^2 = \| \alpha \underline{u} \|_2^2 + \| \underline{v} \|_2^2, \quad \forall \alpha \in \mathbb{C}$$

This is equivalent to

$$\operatorname{Re} \left(\alpha \int_{-\infty}^{+\infty} u(t) v^*(t) dt \right) = 0$$

$\Downarrow$

$$\int_{\mathbb{R}} u(t) v^*(t) dt = 0$$

(because if $z \in \mathbb{C}$ and $\operatorname{Re}(\alpha z) \underset{=0}{\neq} 0 \forall \alpha \in \mathbb{C} \Rightarrow z = 0$)

So here comes the definition:

DEFINITION 4.5.1 (Orthogonal Signals in L_2) The

signals $\underline{u}, \underline{v} \in L_2$ are said to be orthogonal if

$$\langle \underline{u}, \underline{v} \rangle = 0.$$

The n -tuple $(u_1, \dots, u_n)$ is orthogonal if

$$\langle u_n, u_m \rangle = 0 \quad \forall n \neq m.$$

① if $\underline{u} \perp \underline{v} \Rightarrow \underline{u} \perp \alpha \underline{v}$

② $\forall \underline{u} \in L_2 \quad \underline{u} \perp 0$ ③ $\underline{u} \perp \underline{v} \Leftrightarrow \underline{v} \perp \underline{u}.$

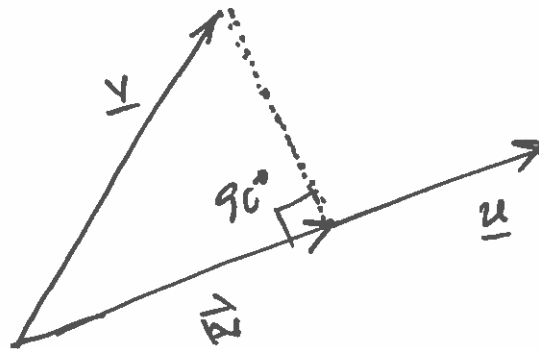
Theorem 4.5.2 (Pythagorean Theorem on n -tuples) (64)

If the n -tuples of vectors $(\underline{u}_1, \dots, \underline{u}_n)$ are $\perp$ in L_2 , then

$$\|\underline{u}_1 + \underline{u}_2 + \dots + \underline{u}_n\|_2^2 = \|\underline{u}_1\|_2^2 + \|\underline{u}_2\|_2^2 + \dots + \|\underline{u}_n\|_2^2$$

PROOF: By induction (Class work)
on n .

PROJECTION: (of one vector onto the other).



The projection of $\underline{v}$ onto $\underline{u}$ is a scaled version of $\underline{u}$ and its length is equal to the length of $\underline{v}$ multiplied by the cosine of the angle between $\underline{u}$ and $\underline{v}$.

$$\underline{w} = (\text{length } \underline{v}) \cos(\angle \text{ between } \underline{v}, \underline{u}) \frac{\underline{u}}{\text{length } \underline{u}} \quad (65)$$

This is the intuition from $\mathbb{R}^2$. But does not work for L_2 because of lack of notion between angles between signals.

An alternate definition of the projection for L_2 is

this. The vector $\underline{w}$ is a projection of the vector $\underline{v}$ onto $\underline{u}$

if $\underline{w}$ is a scaled version of $\underline{u}$ AND $\underline{v} - \underline{w}$ is orthogonal to $\underline{u}$.

DEFINITION 4.5.3: (PROJECTION OF A SIGNAL L_2 onto

another. $\underline{w} = \text{proj}_{\underline{u}} \underline{v}$ iff

(i) $\underline{w} = \alpha \underline{u}$

(ii) $\underline{v} - \underline{w} \perp \underline{u}$.

This is a definition of $\text{proj}_{\underline{u}} v$. Does it exist?

We show by explicit construction that it does.